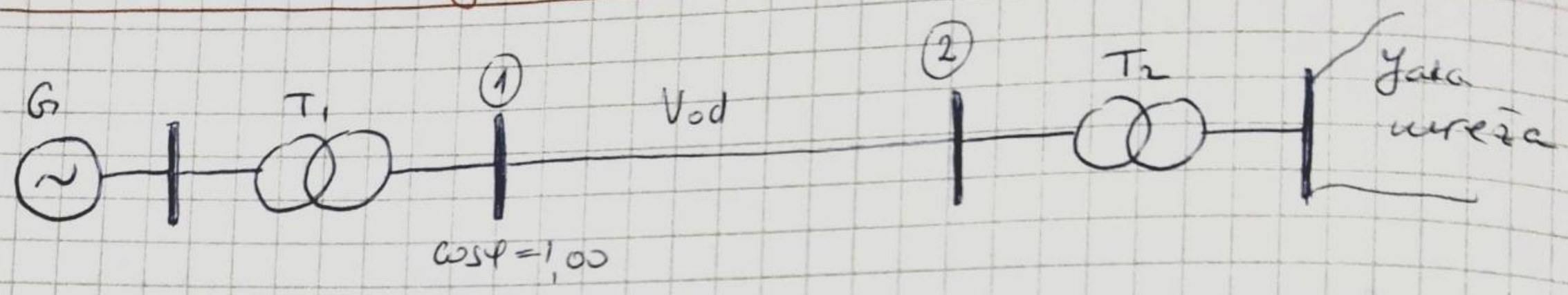
Zadatak za vježbu broj 6



$S_{nG} = S_{nT1} = 500\,MVA$ $\quad U_1 = U_{nV} = 380\,kV$ $\quad S_{nT2} = 500\,MVA$ $\quad U_\infty = const.$

$U_{nG} = 15{,}75\,kV$ $\quad Z_c = 320\,\Omega$ $\quad u_{T2} = 380/220\,kV/kV$ $\quad f = const.$

$u_{T1} = 15{,}75/400\,kV/kV$ $\quad L_V = ?$ $\quad x_{T2} = 10\%$

$x_{dG} = 160\%$

$x_{T1} = 15\%$

Proračun parametara: (380 kV)

$$x_{dG} = \frac{160\%}{100} \cdot \frac{400^2}{500} = 512\,\Omega$$

$$x_{T1} = 0{,}15 \cdot \frac{400^2}{500} = 48\,\Omega$$

$$x_{T2} = 0{,}1 \cdot \frac{380^2}{500} = 28{,}88\,\Omega$$

$Z_c = \sqrt{\frac{l}{c}}$ // karakteristična impedansa

l i c - podužna induktivnost i kapacitivnost

$\beta = \omega\sqrt{lc}$ // fazna konstanta -> mjera promjene faznog ugla dva fazora U ili I duž voda

$\beta = 0{,}06\,\frac{°}{km}$

a) $L_V = ?$

Da bi sistem ostao stabilan na male promenljive:

$$P_{syn} = \frac{\partial P_{Gi}}{\partial \delta_i} > 0$$

$$P_{Gi}' = G_{ii} E_i^2 + \sum_{\substack{k=1 \\ k \neq i}}^{n_G} E_i E_k \left[\cancel{G_{ik}}^{\,0} \cos(\delta_i - \delta_k) + B_{ik} \sin(\delta_i - \delta_k) \right]$$

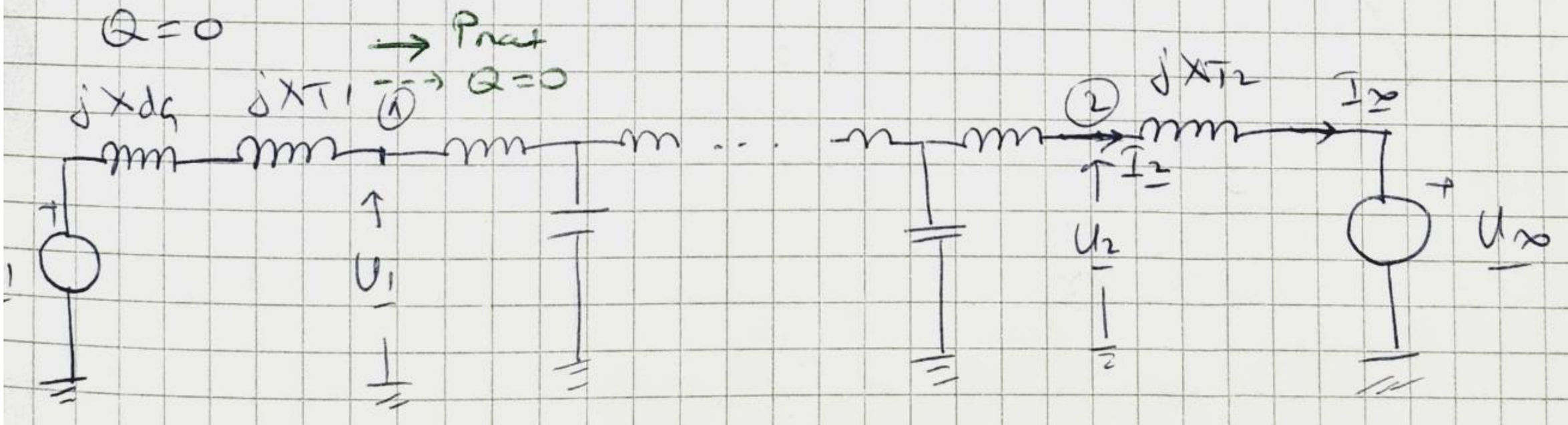
$$P_{G1}' = G_{11} E_1^2 + E_1 U_\infty B_{1\infty} \sin(\delta_1 - \delta_\infty)$$

$$\frac{\partial P_{G1}'}{\partial \delta_1} = E_1 U_\infty B_{1\infty} \cos(\delta_1 - \delta_\infty) > 0$$

$$\delta_1 - \delta_\infty < 90^\circ$$

δ_1 i δ_∞ su fazni stavovi $\underline{E}_1$ i $\underline{U}_\infty$ u odnosu na referentnu osu.

$$P_{nat} = \frac{U_{nv}^2}{Z_c} = \frac{(380\ kV)^2}{320\ \Omega} = 451{,}25\ MW$$



Neka je $\underline{U}_1 = \underline{U}_1^* = U_1 = 380\ kV$

$$-\underline{E}_1 + \underline{I}_1 j(X_{dG} + X_{T1}) + \underline{U}_1 = 0$$

$$\underline{E}_1 = U_1 + j X_{GT1} \cdot \underline{I}_1 \qquad \underline{I}_1 = \frac{S_1^*}{U_1} = \frac{P_{nat} - \cancel{jQ}^{\,0}}{U_1}$$

$$\underline{E}_1 = U_1 + j(X_{dG} + X_{T1}) \cdot \frac{P_{nat}}{U_1} = 380\ kV + j \cdot 560\ \Omega \cdot \frac{451{,}25\ MW}{380\ kV}$$

$\underline{E}_1 = 380\,kV + j\,665\,kV$

$\underline{E}_1 = 765{,}914\,kV \angle 60{,}255^\circ$

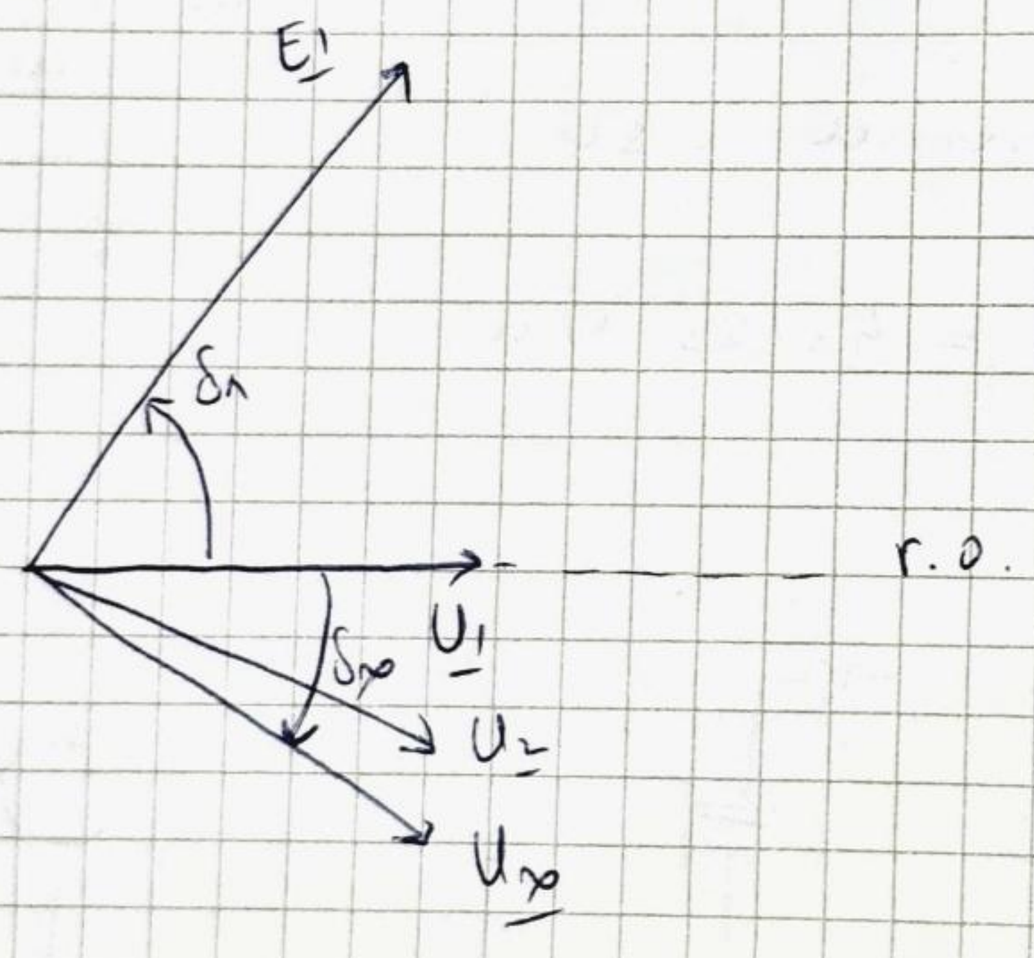
$\underline{U}_1 = \underline{U}_2 \cos\beta L_w + j Z_c \underline{I}_2 \sin\beta L_w$

$\underline{I}_1 = j \frac{\underline{U}_2}{Z_c} \sin\beta L_w + \underline{I}_2 \cos\beta L_w$

$P = P_{nat} \Rightarrow Z_c \underline{I}_2 = \underline{U}_2$

$\underline{U}_1 = \underline{U}_2 (\cos\beta L_w + j \sin\beta L_w) = \underline{U}_2 \cdot e^{j\beta L_w}$, $|\underline{U}_1| = |\underline{U}_2|$

$\underline{I}_1 = j \underline{I}_2 \sin\beta L_w + \underline{I}_2 \cos\beta L_w = \underline{I}_2 e^{j\beta L_w}$



$-\underline{U}_2 + \underline{I}_2 \cdot j X_{T2} + \underline{U}_\infty = 0$

$\underline{U}_\infty = \underline{U}_2 - j X_{T2} \underline{I}_2 = \underline{U}_1 e^{-j\beta L_w} - j X_{T2} \underline{I}_1 e^{-j\beta L_w} =$

$= (U_1 - j X_{T2} I_1) e^{-j\beta L_w} = (380 - j \cdot 34{,}295) e^{-j\beta L_w}\,kV =$

$= 381{,}544 \angle -(5{,}157^\circ + \beta L_w)$

$$\delta_1 - \delta_\infty = 60{,}255^\circ + 5{,}157^\circ + \beta L_v < 90^\circ$$

$$9{,}06 \frac{^\circ}{km} \cdot L_v < 24{,}588^\circ$$

$$\boxed{L_v < 409{,}8 \text{ km}}$$

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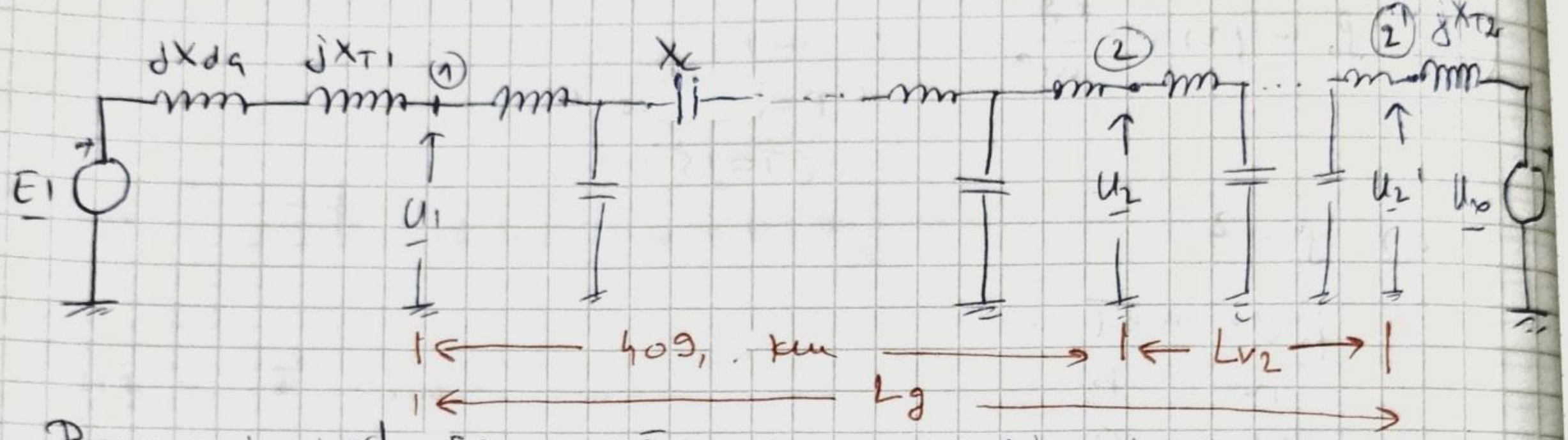
b) Priključivanjem baterije kondenzatora, tj. kompenzacijom $\frac{1}{2}$ reaktanse voda iz dijela pod a) postiže se mogućnost prenosa iste snage na veću udaljenost a da pri tom sistem ostane stabilan.



Prenosni vod se može posmatrati kao kaskadna veza dva voda.

1. vod = kompenzovani, dužine $L_{v1} = 409{,}8$ km.

Kompenzovano je 50% reaktanse voda pa je podužna induktivnost upola manja od podužne induktivnosti prije kompenzacije.

Podužna induktivnost figuriše u izrazu za Z_c i β pa vrijedi:

$$Z_{c_1} = \sqrt{\frac{l'}{c}} = \frac{Z_c}{\sqrt{2}} = 226{,}274\,\Omega$$

$$\beta_1 = \sqrt{l'c} = \frac{\beta}{\sqrt{2}} = 0{,}04243\ \frac{°}{km}$$

2. vod: ima podužne parametre nekompenzovanog voda, $Z_{c2} = 320\,\Omega$, $\beta = 0{,}06\ \frac{°}{km}$

Dužina voda 2 je jednaka dodatnoj dužini na koju se snaga može prenijeti.

$P = 451{,}25\ MW \neq P_{nat\,1}$

$$\underline{U}_1 = \underline{U}_2 \cos \beta_1 L_{v1} + j Z_{c1} \underline{I}_2 \sin \beta_1 L_{v1}$$

$$\underline{I}_1 = j \frac{\underline{U}_2}{Z_{c1}} \sin \beta_1 L_{v1} + \underline{I}_2 \cos \beta_1 L_{v1}$$

$$\begin{bmatrix} \underline{U}_1 \\ \underline{I}_1 \end{bmatrix} = \begin{bmatrix} \cos \beta_1 L_{v1} & j Z_{c1} \sin \beta_1 L_{v1} \\ j \frac{1}{Z_{c1}} \sin \beta_1 L_{v1} & \cos \beta_1 L_{v1} \end{bmatrix} \begin{bmatrix} \underline{U}_2 \\ \underline{I}_2 \end{bmatrix}$$

$$\begin{bmatrix} \underline{U}_2 \\ \underline{I}_2 \end{bmatrix} = \begin{bmatrix} \cos \beta_1 L_{v1} & -j Z_{c1} \sin \beta_1 L_{v1} \\ -j \frac{1}{Z_{c1}} \sin \beta_1 L_{v1} & \cos \beta_1 L_{v1} \end{bmatrix} \begin{bmatrix} \underline{U}_1 \\ \underline{I}_1 \end{bmatrix}$$

$$\underline{U}_2 = \cos \beta_1 L_{v1} \underline{U}_1 - j Z_{c1} \sin \beta_1 L_{v1} \underline{I}_1$$

$$\underline{I}_2 = -j \frac{1}{Z_{c1}} \sin \beta_1 L_{v1} \underline{U}_1 + \cos \beta_1 L_{v1} \underline{I}_1$$

$$\underline{U}_1 = U_1 = 380\ kV$$

$$I_1 = \frac{P}{U_1} = 1187{,}5\ A$$

$$\underline{U}_2 = \cos(17{,}386^\circ) \cdot 380\ kV - j \cdot 226{,}274 \cdot \sin(17{,}386^\circ)\, 1187{,}5\ A$$

$$\underline{U}_2 = (362{,}64 - j\,80{,}29)\ kV = 371{,}42\ kV \angle -12{,}484^\circ$$

$$\underline{I}_2 = -j \frac{380\ kV}{226{,}274\,\Omega} \cdot \sin(17{,}386^\circ) + \cos(17{,}386^\circ) \cdot 1187{,}5\ A$$

$$\underline{I}_2 = (1133{,}247 - j\,501{,}811)\ A = 1239{,}38\ A \angle -23{,}88^\circ$$

$$\underline{E}_1 = 765{,}914\ kV \angle 60{,}255^\circ$$

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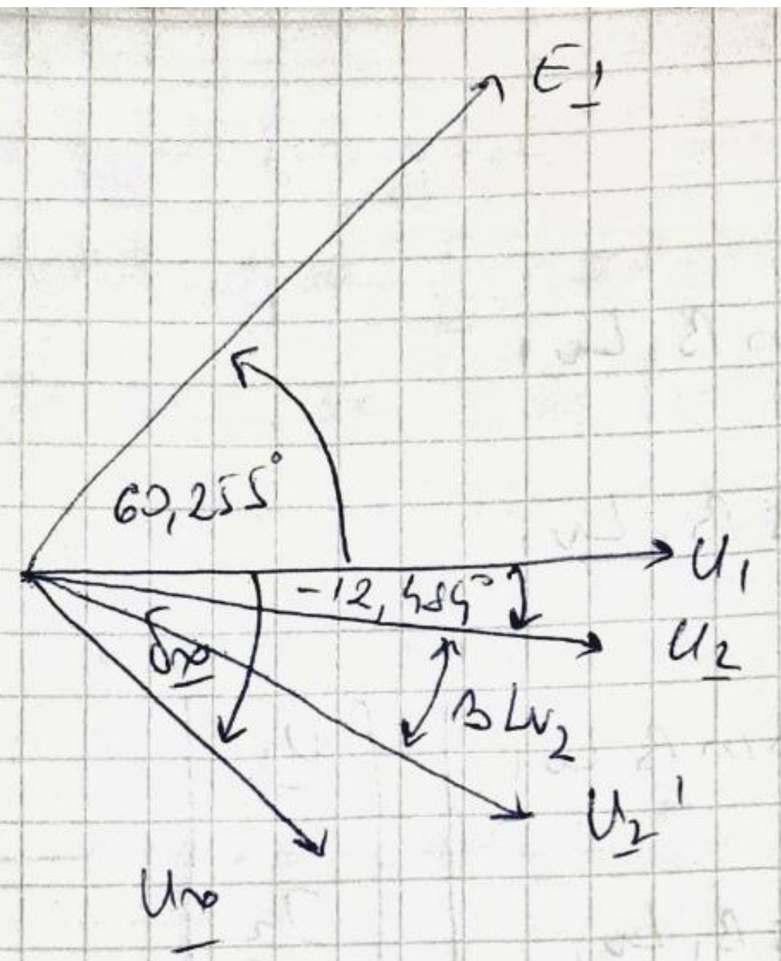

Za drugi dio voda (duzine L_{v2}) $P = 451,25\ MW$ je prirodna snaga, pa vrijedi: $\underline{U}_2' = Z_c \underline{I}_2'$

$$\underline{U}_2 = \underline{U}_2' \cos \beta_2 L_{v2} + j Z_{c2} \underline{I}_2' \sin \beta_2 L_{v2}$$

$$\underline{I}_2 = j \frac{\underline{U}_2'}{Z_{c2}} \sin \beta_2 L_{v2} + \underline{I}_2 \cos \beta_2 L_{v2}$$

$$\underline{U}_2 = \underline{U}_2' (\cos \beta L_{v2} + j \sin \beta L_{v2}) = \underline{U}_2' e^{j \beta L_{v2}}$$

$$\underline{I}_2 = \underline{I}_2' (j \sin \beta L_{v2} + \cos \beta L_{v2}) = \underline{I}_2' e^{j \beta L_{v2}}$$

$$\underline{U}_2' = \underline{U}_2 e^{-j \beta L_{v2}} = 371,42\ kV \angle -(\beta L_{v2} + 12,484°)$$

$$\underline{I}_2' = \underline{I}_2 e^{-j \beta L_{v2}} = 1239,38\ A \angle -(\beta L_{v2} + 23,88°)$$

$$-\underline{U}_2' + j X_{T2} \underline{I}_2' + \underline{U}_{\infty} = 0$$

$$\underline{U}_{\infty} = \underline{U}_2' - j X_{T2} \underline{I}_2'$$

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$$\underline{U}_\infty = 371{,}42\,kV \cdot e^{-j(\beta L_2 + 12{,}484^\circ)} \qquad -j(\beta L_2 + 23{,}88^\circ)$$

$$-j \cdot 28{,}88 \cdot 1239{,}38 \cdot e$$

$$\underline{U}_\infty = 371{,}42\left(\cos(\beta L_2 + 12{,}484^\circ) - j\sin(\beta L_2 + 12{,}484^\circ)\right)$$

$$-j \cdot 35{,}793\left(\cos(\beta L_2 + 23{,}88^\circ) - j\sin(\beta L_2 + 23{,}88^\circ)\right)$$

$$\underline{U}_\infty = 371{,}42\cos(\beta L_2 + 12{,}484^\circ) - 35{,}793\sin(\beta L_2 + 23{,}88^\circ)$$

$$-j\left(371{,}42\sin(\beta L_2 + 12{,}484^\circ) + 35{,}793\cos(\beta L_2 + 23{,}88^\circ)\right)$$

$$\delta_\infty = -\operatorname{arctg}\frac{371{,}42\sin(\beta L_2 + 12{,}484^\circ) + 35{,}793\cos(\beta L_2 + 23{,}88^\circ)}{371{,}42\cos(\beta L_2 + 12{,}484^\circ) - 35{,}793\sin(\beta L_2 + 23{,}88^\circ)}$$

$$\delta_1 - \delta_\infty < 90^\circ$$

$$\delta_\infty > 60{,}255^\circ - 90^\circ$$

$$\delta_\infty > -29{,}745^\circ$$

Granični slučaj: $\delta_\infty = -29{,}745^\circ$

$$\frac{371{,}42\sin(\beta L_2 + 12{,}484^\circ) + 35{,}793\cos(\beta L_2 + 23{,}88^\circ)}{371{,}42\cos(\beta L_2 + 12{,}484^\circ) - 35{,}793\sin(\beta L_2 + 23{,}88^\circ)} = -\operatorname{tg}\delta_\infty$$

$$371{,}42\left(\sin\beta L_2\cos 12{,}484^\circ + \sin 12{,}484^\circ\cos\beta L_2\right)$$

$$+35{,}793\left(\cos\beta L_2\cos 23{,}88^\circ - \sin\beta L\sin 23{,}88^\circ\right) =$$

$$= 0{,}57143\Big[371{,}42\left(\cos\beta L_2\cos 12{,}484^\circ - \sin\beta L_2\sin 12{,}484^\circ\right)$$

$$-35{,}793\left(\sin\beta L_2\cos 23{,}88^\circ + \sin 23{,}88^\circ\cos\beta L_2\right)\Big]$$

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$$362{,}64 \sin \beta L_{v2} + 80{,}288 \cos \beta L_{v2} + 3273 \cos \beta L_{v2} -$$

$$-14{,}49 \sin \beta L_{v2} = 207{,}22 \cos \beta L_{v2} - 45{,}88 \sin \beta L_{v2}$$

$$-18{,}7 \sin \beta L_{v2} - 8{,}28 \cos \beta L_{v}$$

$$375{,}33 \sin \beta L_{v2} = 85{,}922 \cos \beta L_{v2}$$

$$tg\, \beta L_{v2} = 0{,}2289$$

$$\beta L_{v2} = 12{,}89^\circ \Rightarrow L_{v2} = \frac{12{,}89^\circ}{0{,}06 \frac{^\circ}{km}}$$

$$L_{v2} = 214{,}9 \text{ km}$$

$$L_g = L_{v1} + L_{v2}$$

$$\boxed{L_g = 624{,}7 \text{ km}}$$

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